

## Lesson 6-3: Explore Numerical Expressions with Exponents

Grade 6 Mathematics · Standard 6.AT.5

Name: \_\_\_\_\_ Date: \_\_\_\_\_

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

### Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

#### Question 1 SELECTED RESPONSE · 1 POINT

What is the value of  $4^3$ ?

- ☐ A 7
- ☐ B 12
- ☐ C 43
- ☐ D 64

#### Question 2 SELECTED RESPONSE · 1 POINT

What is the value of  $6^2$ ?

- ☐ A 8
- ☐ B 12
- ☐ C 36
- ☐ D 62

#### Question 3 SELECTED RESPONSE · 1 POINT

Which expression shows  $5^3$  as repeated multiplication?

- ☐ A  $3 \times 3 \times 3 \times 3 \times 3$
- ☐ B  $5 \times 5 \times 5$
- ☐ C  $5 \times 3$
- ☐ D  $5 + 5 + 5$

**Question 4**    SELECTED RESPONSE · 1 POINT

Evaluate:  $3 \times (4 + 2)^2$

- ☐ A 36
- ☐ B 54
- ☐ C 108
- ☐ D 144

**Question 5**    SELECTED RESPONSE · 1 POINT

The drum loop plays  $3^4$  beats. What is the value of  $3^4$ ?

- ☐ A 7
- ☐ B 12
- ☐ C 27
- ☐ D 81

**Question 6**    SELECTED RESPONSE · 1 POINT

A synth pattern repeats  $7 \times 7 \times 7$ . How do you write that as a power?

- ☐ A  $3^7$
- ☐ B  $7^3$
- ☐ C  $7 \times 3$
- ☐ D  $21^3$

**Part B — Test-Format Questions**

These match the state test's formats: two-part questions, select-all, and written responses.

**Question 7****TWO-PART QUESTION****Part A**

Which value is equal to the power  $3^4$ ?

*Fill in the bubble next to the one best answer.*

- ☐ A 7
- ☐ B 12
- ☐ C 64
- ☐ D 81

**Part B**

*Answer Part A before Part B — Part B asks about your reasoning.*

Which reasoning shows why your answer to Part A is correct?

- ☐ A Multiply the base by the exponent:  $3 \times 4 = 12$ .
- ☐ B Add the base and the exponent:  $3 + 4 = 7$ .
- ☐ C The exponent 4 tells how many times to use the base 3 as a factor:  $3 \times 3 \times 3 \times 3 = 81$ .
- ☐ D Use the exponent as the base and the base as the exponent:  $4 \times 4 \times 4 = 64$ .

**Question 8****SELECT ALL THAT APPLY**

Select ALL statements that are true about the power  $2^5$ .

*Mark the box next to EVERY answer that is true. More than one is correct.*

- ☐ A The base is 2.
- ☐ B The exponent is 5.
- ☐ C It means  $2 \times 2 \times 2 \times 2 \times 2$ .
- ☐ D Its value is 10.
- ☐ E Its value is 32.
- ☐ F It means  $5 \times 5$ .

**Part C — Show Your Thinking**

Answer in writing, the way the state test's constructed responses work.

**Question 9****WRITTEN RESPONSE · 2 POINTS**

A bacteria colony doubles every hour. Starting with 1 bacterium, write a power expression for the number after 6 hours. Explain why exponential growth is so much faster than adding the same number each hour.

*Write your answer in complete sentences. Show or explain your mathematical thinking.*

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When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.